



Technology Stack

Zero-Hardware

Mobile & Frontend	React Native / React 19, Tailwind CSS v4, Lucide-React, MapLibre GL, Sonner
Backend & APIs	Python 3.11, FastAPI (Async), Uvicorn, WebSockets (WSS), Node.js Gateway
AI & Graph Deep Learning	PyTorch 2.5, PyTorch Geometric (PyG 2.6), ONNX Runtime
Uncertainty Calibration	Inductive Conformal Prediction
Physics Kinematics	Newton-Davis Equation, Runge-Kutta 4th-Order (RK4)
Spatial & Time-Series DB	PostgreSQL 16, PostGIS (1D LRS), TimescaleDB
Streaming & In-Memory Cache	Apache Kafka (KRaft), Redis 7.2 (Pub/Sub)

Statutory Data Ingestion

BEL RTIS (ISRO NavIC), CRIS COA, e-Caution (TSR), Open-Meteo

DevOps & Security

Docker, Kubernetes (K8s), OAuth 2.0 / JWT, AES-256 GCM

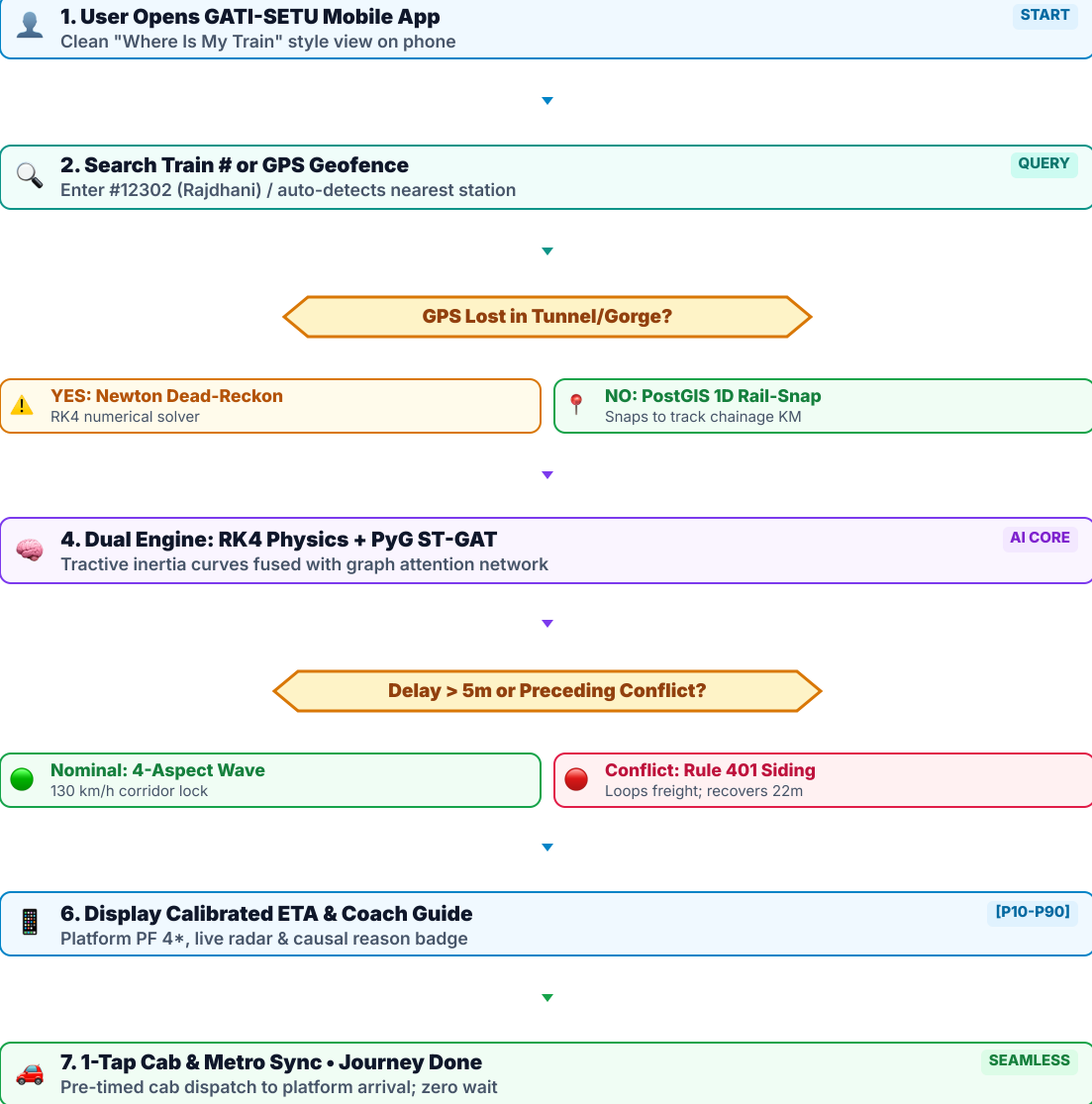
100% ZERO-HARDWARE

Taps statutory Indian Railways GPS & CRIS feeds. Zero engine retrofits.



End-to-End User & System Flow

Traveler Journey



Working Prototype & Capabilities

Verified Live



Live Passenger App

PostGIS 1D LRS & Kinematics

Snaps 2D GPS to 1D track chainage KM; RK4 ODE solver computes acceleration.

$a(t) = [F_{trac}(v) - R_{davis} - F_{grade}] / M_{eff}$

PyG ST-GAT Corridor Graph

Multi-head graph attention models downstream ripple delays across stations up to 150 km.

$a_{ij} = \text{Softmax}(\text{LeakyReLU}(a^T [Wh_i || Wh_j]))$

Inductive Conformal Uncertainty

Certified 90% non-parametric confidence window (23:40–23:46) immune to fog halts.

Section Controller & Station CIDS

Rule 401 Siding Engine loops freight (+22m recovery) & syncs platform LED/audio displays.

MASTER IMPLEMENTATION ARCHITECTURE (INPUT STAGE • PROCESSING STAGE • OUTPUT STAGE)

End-to-End Operational Lifecycle

